The Grounding Paradox: Why True Physical Inputs Make a Solubility Model Worse, and How to Measure It

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Abstract

Physics-informed models for solid–liquid solubility factor the target $\ln x_2 = -\Phi(T) - \ln \gamma_2$ into a pure-component crystal term and a pair activity term, and increasingly compute the activity term with a *differentiable thermodynamic closure*—here COSMO-SAC over a network-predicted σ-profile—on the premise that grounding the closure on true physical inputs improves it. It does the opposite: replacing the model's learned σ-profiles with the true, externally measured ones (VT-2005) makes both activity and solubility prediction *worse* than a free learned head, an effect we confirm three ways. We turn this paradox into a measurement. Because the physical input has external ground truth, the true-input error splits—with no model fit—into a part the inputs cannot resolve and a part the fixed closure gets wrong; on infinite-dilution activity data the closure is the binding term, and a downstream recalibration recovers *calibration* but not *accuracy*. The failure is not incidental. The physics-informed model does not beat a matched black box on accuracy at *any* training-data budget—including the low-data regime where an inductive bias should help most—nor on temperature extrapolation, and we prove why: the crystal/activity split is non-identifiable from solubility, and the closure is misspecified where we can measure it. The paper's contribution is therefore not accuracy but a reproducible, proved method for locating where a physics prior's ceiling lies, and—for this system—an account of why more physics does not help; what the prior does buy is an interpretable decomposition and calibrated uncertainty, not lower error.

1 Introduction

Coupling a graph encoder to an explicit thermodynamic bottleneck is a common recipe for out-of-distribution solubility prediction: the model must route its prediction through interpretable physical factors, and each factor can in principle be grounded on abundant single-component data that a black-box predictor cannot exploit. The expectation is that the closer the model's physical inputs are to the truth, the better it should predict. On the activity branch this expectation fails. We characterize the failure quantitatively, and it is the characterization, more than the failure itself, that this paper is about. That true σ-profiles can hurt a COSMO-SAC pipeline will not surprise a thermodynamics specialist who knows the model's limits; the contribution is not the failure but an externally grounded procedure—demonstrated here on one closure and one low-γ regime—that *measures* where such a physics prior's ceiling lies and attributes it to the closure rather than the inputs. Whether the finding of closure-dominance generalizes beyond this regime we leave as a conjecture (§7).

Contributions.

- **The grounding paradox (empirical, triple-confirmed).** Substituting true VT-2005 σ-profiles into the differentiable COSMO-SAC closure degrades activity ($\ln \gamma_2^\infty$) and solubility ($\ln x_2$) prediction relative to a free learned activity head, the opposite of the intended benefit (§6.1).
- **A measurement affordance, not a theorem.** Because the physical latent $z^\star$ (the true σ-profile) has external ground truth, the true-input excess risk splits *exactly* as $B = B_{\text{closure}} + B_{\text{insuff}}$ with no model fit (Lemma 3). The decomposition and its estimators are textbook; the new element is exploiting an externally measured latent to bound the split empirically. On the convention-robust evidence—a convention-independent bound and the more-physics-worse effect—B_{closure} (misspecified decoder) exceeds B_{insuff} (input insufficiency): the ceiling is the closure (§6.2).
- **A negative actionability result (Gate B).** A post-hoc output-space residual placed on the physics prediction recovers calibration but not accuracy; the closure binding cannot be undone downstream (§6.4).
- **An exhaustive, explained negative.** The physics prior does not beat a matched black box on accuracy at any training-data budget—including the low-data regime where an inductive bias should help most (§6.7)—nor on temperature, nor via crystal grounding; read against the lemmas and the closure measurement, this locates the ceiling as *structural*, not incidental (§7). We make no accuracy claim.
- **Framework and honest scoping.** We prove three elementary structural lemmas (non-identifiability, class-dependent information, and the closure/insufficiency split; Lemmas 1–3, App. B), read the deeper theory against cited results on model discrepancy and calibration, and ship reproducible diagnostics.

2 Related work

Two families of methods now dominate solid solubility and solvation prediction, and they make opposite bets about physical structure. Physics-informed pipelines such as SolProp [15] assemble a thermodynamic cycle from separately learned components (solvation free energy, vapor pressure, fusion properties), so that each prediction is decomposable and the intermediate quantities carry physical meaning. Direct models forgo that structure: FastSolv [1] regresses solubility from molecular descriptors with a descriptor-based deep neural network (fastprop), and it is the current leader on new-solute extrapolation, sitting near the aleatoric floor set by inter-laboratory disagreement. The ordering is unambiguous on accuracy: the black box leads, and the physics-informed cycle trails it. Our work does not try to overturn this ordering. It asks instead where the ceiling of the physics prior comes from, and it uses a matched direct control that shares our encoder so that the accuracy gap is attributable to the closure rather than to representation.

The activity coefficient is the term our closure must supply, and learned surrogates for it have advanced quickly. Winter et al. [18] predict infinite-dilution activity coefficients $\ln \gamma_2^\infty$ directly from SMILES with a language model, and Sanchez-Medina et al. [12] embed a Gibbs-Helmholtz constraint in a graph network to capture the temperature dependence of $\ln \gamma_2^\infty$. These treat $\ln \gamma_2$ as the regression target. We take the complementary route of retaining a mechanistic activity model, either COSMO-SAC over a predicted σ-profile or NRTL, as a differentiable closure inside the solver, so that the network predicts physical inputs rather than the coefficient itself. That choice is what exposes the failure mode we measure: the misspecification of a fixed mechanistic map can dominate the insufficiency of its learned inputs.

We borrow several diagnostic lenses and claim no new theory from them. The gap between a fixed physical map and reality is the model-discrepancy problem of Kennedy and O'Hagan [7] and its calibration critique [2]; separating whether predictions are right on average from whether they are sharp is the calibration refinement and reliability-resolution decomposition of Murphy and DeGroot [10, 4]; fitting a wrong model still converges to a well-defined pseudo-true limit under the quasi-maximum-likelihood account of White [17]. That the decomposition is only weakly constrained by solubility alone is a matter of parameter sloppiness [13, 5] and of structural versus practical identifiability [11, 16]. Finally, the risk that apparent skill reflects pipeline choices rather than physics is the concern of underspecification [3], leakage and reproducibility [6], and compositional risk [9]. Our data come from BigSolDB 2.0 [8].

Positioned against this literature, our contribution is not a new model and not a new theorem. It is a measurement. We quantify where a physics prior's accuracy ceiling actually lives by decomposing the error into a closure-misspecification part B_{closure} and an input-insufficiency part B_{insuff}, and we show that feeding true physical inputs through the closure can make predictions worse when B_{closure} dominates. What the physics buys, then, is not accuracy but an interpretable decomposition, an affordance for measuring its own limiting term, an honest map of where it applies, and calibrated uncertainty, while FastSolv and SolProp remain the reference points for accuracy.

3 Problem setup and notation

For a solute s in solvent v at temperature T, the target is $y = \ln x_2$, the log saturated mole fraction. Solid–liquid equilibrium gives

$$\ln x_2 = -\underbrace{\frac{\Delta H_{\text{fus}}}{R}\Big(\tfrac{1}{T} - \tfrac{1}{T_m}\Big)}_{\Phi(T)\ \text{(crystal, depends on } s\text{)}} - \underbrace{\ln \gamma_2(x_2, T)}_{\text{activity, depends on } (s,v)}, \tag{1}$$

with the constant-ΔC_p correction omitted for clarity. The activity term is produced by a *differentiable closure* g: a graph encoder predicts a σ-profile for each molecule, and COSMO-SAC maps the pair of profiles to $\ln \gamma_2$. Write $z^\star$ for the *true* physical latent, the externally measured σ-profile (VT-2005 database), and $g(z^\star)$ for the closure's output when fed the true profile. The model's own head instead predicts a *learned* profile $\hat{z}$ and evaluates $g(\hat{z})$; a matched *DirectGNN* control shares the encoder but emits $\ln x_2$ with no closure and is the reference for "what the physics bottleneck costs" (Fig. 1).

The grounding intervention. Feeding $z^\star$ in place of $\hat{z}$ (a σ-oracle) isolates the closure from encoder error: if the closure were well specified, $g(z^\star)$ should be the best available activity estimate. The paradox is that it is not.

Data. Solubility measurements are a Bemis–Murcko scaffold split of BigSolDB [8], so test solutes are unseen at training; the activity labels m are experimental infinite-dilution coefficients $\ln \gamma_2^\infty$ compiled from ThermoML, and the true σ-profiles $z^\star$ are the VT-2005 database.

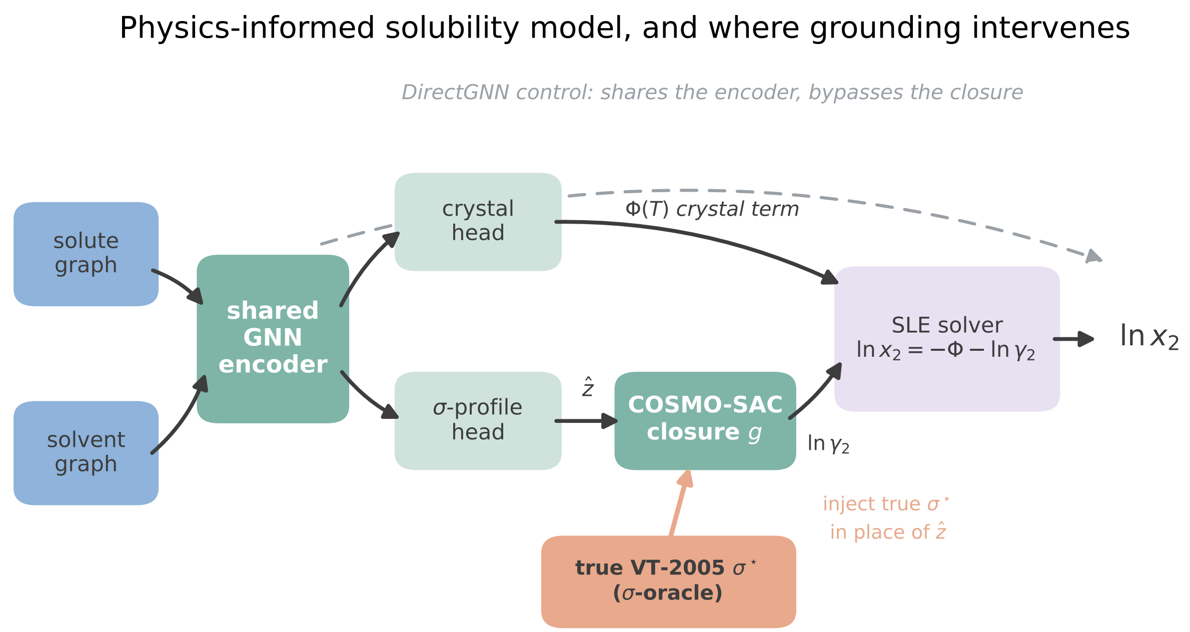

Figure 1: The model and the grounding intervention. Solute and solvent graphs pass through a shared encoder to a crystal head (giving the ideal term $\Phi(T)$) and a σ-profile head; the predicted profile $\hat{z}$ feeds the COSMO-SAC closure g, which returns $\ln \gamma_2$, and an SLE solver combines the two into $\ln x_2$. The σ-oracle replaces $\hat{z}$ with the true VT-2005 profile $z^\star$ (salmon). DirectGNN shares the encoder but bypasses the closure.

4 Theory: locating a physics prior's ceiling

4.1 Why a free head can compensate: structural non-identifiability

The crystal/activity split is not identifiable from solubility data alone, which is precisely why an unconstrained head can silently absorb crystal error into the activity term.

Lemma 1 (Structural non-identifiability of the SLE split)**.** *Consider the infinite-dilution SLE observable $\mu(T) = -\Phi(T) - \ln \gamma_2^\infty(T)$ with $\Delta C_p = 0$, so that $\Phi(T) = (\Delta H_{\text{fus}}/R)(1/T - 1/T_m)$ is affine in $u = 1/T$, and a symmetric two-suffix activity class ($\ln \gamma_2 = \ln \gamma_2^\infty (1 - x_2)^2$) in which $\ln \gamma_2^\infty(T)$ is affine in u. Then, for any design with at least two distinct temperatures, μ depends on the parameters only through the two-dimensional statistic (its slope and intercept in u), and the activity intercept and slope already span that plane; hence the crystal parameters $(\Delta H_{\text{fus}}, T_m)$ are unidentified, the profiled Fisher information for ΔH_{fus} is exactly zero, and the indeterminacy set is two-dimensional. The deficiency is closed only by a rank-completing external constraint (a direct crystal label, or a fixed activity slope). At finite dilution the composition factor $(1 - x_2)^2$ breaks the exact affinity, leaving a non-zero but near-degenerate information of order x_2^2 (numerically, condition number 2.3×10^5; §6.5).*

This is the standard structural-identifiability picture for a collinear design [11, 16]; we prove it in the SLE form (App. B) and audit it numerically in §6.5. It is *enabling background*, not our headline: it explains why grounding an under-determined factor on external single-component data is necessary, and why compensation between the factors is expected. By itself it says nothing about whether grounding helps. That depends on whether the closure that consumes the grounded inputs is correctly specified, which is the question we make measurable next.

The non-identifiability is not binary but graded by how much of the crystal direction the activity class can absorb.

Lemma 2 (Class-dependent efficient information)**.** *Let the activity nuisance lie in a class $\mathcal{H}$ with tangent projection $\Pi_{\mathcal{H}}$, and let $v = \partial\mu/\partial\Delta H_{\text{fus}}$ be the crystal score. The efficient information for ΔH_{fus} is $\mathcal{I}_{\text{eff}}(\Delta H_{\text{fus}}) = \sigma^{-2}\,\|v - \Pi_{\mathcal{H}}v\|^2$. If $\mathcal{H}$ contains the crystal score's span $\{1, 1/T\}$ (a free intercept and slope) then $\Pi_{\mathcal{H}}v = v$ and $\mathcal{I}_{\text{eff}} = 0$, recovering Lemma 1; if $\mathcal{H}$ is restricted (the slope is known or constrained) then $\mathcal{I}_{\text{eff}} > 0$ and the Cramér–Rao bound on ΔH_{fus} is finite.*

The proof is the efficient-score construction (App. B): project the crystal score onto the orthogonal complement of the nuisance tangent space; the squared norm of the residual is the efficient information. The numerical audit of §6.5 quantifies both regimes, and shows the honest corollary: merely *restricting* the activity slope leaves $\text{sd}(\Delta H_{\text{fus}}) \approx 8{,}251$ J/mol — finite but unusable — whereas an *external* crystal label, which contributes a score direction outside the activity tangent space, closes it to $\approx 1{,}689$ J/mol.

4.2 The closure–insufficiency decomposition

Let m be the activity target (experimental $\ln\gamma_2^\infty$) and $g(z^\star)$ the closure evaluated on the true latent. The excess risk of the true-input path over the Bayes predictor decomposes exactly.

Lemma 3 (Exact closure–insufficiency decomposition and its one-sided bounds)**.** *For a* fixed *(non-fitted) closure g,*

$$\underbrace{\mathbb{E}\big[(m - g(z^\star))^2\big]}_{\textit{MSE of the true-input path}} \quad = \quad \underbrace{\mathbb{E}\big[\mathrm{Var}(m \mid z^\star)\big]}_{B_{\text{insuff}}\ \textit{(inputs cannot resolve)}} \quad + \underbrace{\mathbb{E}\big[(\mathbb{E}[m \mid z^\star] - g(z^\star))^2\big]}_{B_{\text{closure}}\ \textit{(decoder mis-map)}}. \tag{2}$$

The cross term vanishes identically because $\mathbb{E}[m \mid z^\star] - g(z^\star)$ is $z^\star$-measurable and $\mathbb{E}[(m - \mathbb{E}[m \mid z^\star]) \mid z^\star] = 0$. Both components are separately estimable from data with no model fit*, because $z^\star$ is externally observed.*

Equation (2) is a bias–variance split of model discrepancy in the sense of Kennedy–O'Hagan [7] and Brynjarsdóttir–O'Hagan [2]: B_{closure} is the systematic discrepancy of a structurally fixed physical map, and its dominance over B_{insuff} is the empirical content of "the ceiling is the closure, not the inputs." We do not claim (2) as new; the affordance is that the external groundedness of $z^\star$ lets us *measure* the split rather than assume it. **The exactness is contingent on g being fixed**: for a *fitted* decoder the cross term does not vanish and a plug-in point decomposition is invalid (we observe a spurious -0.36 cross term when $\mathbb{E}[m \mid z^\star]$ is replaced by a fitted estimate). We therefore report *one-sided bounds*, not a point split (each proved in App. B):

- $B_{\text{closure}} \geq (\mathbb{E}[m] - \mathbb{E}[g])^2$, *assumption-free in smoothness* (Jensen; a constant decoder offset, which cannot be input insufficiency)—it does assume m and g are on the same $\ln\gamma_2^\infty$ reference state, which we check directly (§6.2).
- $B_{\text{insuff}} \leq \mathbb{E}[\mathrm{Var}(m \mid \text{bin}(g(z^\star)))]$, a valid *upper* bound (law of total variance, since a binning of $g(z^\star)$ coarsens $z^\star$).

Because $B_{\text{insuff}} = \mathbb{E}[\mathrm{Var}(m \mid z^\star)]$ depends only on $(m, z^\star)$, it is *independent of the closure convention*: any disagreement between closure variants is decoder disagreement and lands in B_{closure}.

4.3 Supporting structural facts

Three further consequences of the same collinear geometry are used informally and are not stated as theorems. *(i) Temperature extrapolation.* Within the affine-in-$1/T$ activity family, same-pair prediction at a temperature $T^\star$ outside the training range is an affine extrapolation whose error is controlled by the fitted-slope variance times $|1/T^\star - \overline{1/T}|$, whereas an unstructured temperature encoder is unconstrained beyond its support. This is the one place the physical *structure* is expected to help; §6.9 reports the class-level margin, which is a property of the hypothesis class and not an accuracy claim for the trained model (indeed the trained models trail the van't Hoff baseline). *(ii) Geometry of compensation.* As the training window $\Delta T \to 0$, $1/T$ becomes nearly constant, the crystal and activity design columns become collinear, and the smallest Fisher eigenvalue tends to zero (the ΔT sweep of §6.5 reaches conditioning 10^{16} at $\Delta T = 10\,\mathrm{K}$), so an optimizer prefers to split error between the factors rather than fit either correctly. *(iii) Grounding design.* Each single-component stream rank-completes a different deficient direction of Lemma 1; this is the design rule behind the auxiliary streams, though (as §6.6 shows for a dose-limited crystal run) it does not by itself promise an end-task gain.

4.4 What we prove, and what we do not claim

We prove the three structural lemmas the measurement rests on — the split's non-identifiability (Lemma 1), its class-dependence (Lemma 2), and the exact closure/insufficiency decomposition with its one-sided bounds (Lemma 3); the proofs are elementary and collected in App. B. We claim no more. In particular there is no "optimal grounding" theorem and no necessary-and-sufficient condition for the decomposition to improve out-of-distribution accuracy — the honest empirical answer (§6) is that it does not, and an earlier draft's conditional-optimality "iff" is refuted by exactly that. The lemmas are read against, not in competition with, known results on model discrepancy [7, 2], parameter sloppiness [13, 5], structural and practical identifiability [11, 16], estimation under misspecification [17], and underspecification [3]; the calibration-versus-accuracy separation of §6.4 is the classical refinement decomposition of a probability forecast [10, 4].

5 Methods

The model is a controlled comparison: a shared graph backbone feeds either an explicit thermodynamic bottleneck (*TGNNSolv*) or a direct regressor (*DirectGNN*) that shares the backbone but emits $\ln x_2$ with no physics. The gap between them measures what the bottleneck costs or buys. This section describes the shared backbone, the three interchangeable activity closures, the solid–liquid-equilibrium (SLE) solver and its implicit differentiation, the grounding intervention, and the training protocol. Exact hyperparameters, the full feature list, and the complete loss taxonomy are in the SI (§A).

5.1 Data, splits, and labels

Solubility measurements are drawn from BigSolDB 2.0 [8] and partitioned by Bemis–Murcko scaffold (`scripts/prepare_data.py`) so that every test solute is unseen at training; row membership is set by a melting-point-independent miscibility/structure filter, and crystal T_m, ΔH_{fus}, Hansen parameters, and infinite-dilution activity coefficients $\ln \gamma_2^\infty$ are merged on as per-solute labels where available. Two data hazards are handled explicitly. Melting-point tables arrive in either °C or Kelvin and are auto-detected by column median (a past double-conversion inflated every T_m by

273 K, which propagates directly into the ideal term); and the seeded scaffold split is not stable across pipeline versions, so every regeneration is treated as a new split with all downstream artifacts recomputed. Crystal supervision is scarce: the test and validation splits contain essentially no measured ΔH_{fus}, which is the quantitative backbone of the identifiability argument (§4). External single-component data used for grounding are an open crystal-property pool (∼15,000 melting points), the VT-2005 σ-profile database, and ThermoML infinite-dilution activity coefficients; each is filtered to exclude any molecule whose scaffold appears in the validation or test split.

5.2 Graph encoder and pair representation

Solute and solvent are encoded by a *shared* graph network with three interchangeable backbones: a message-passing network (MPNN), a hybrid local-message/global-attention network (GPS), and a physics-channelled variant (TIMP) that splits messages into dispersive, polar, and hydrogen-bonding channels supervised toward Hansen parameters to prevent channel collapse. Small hydride solvents such as water are given explicit hydrogens so the polar and hydrogen-bond channels activate. Solute and solvent embeddings interact through a bipartite cross-attention/message-passing block and a physics-aware readout, and are combined into a pair representation $[\, g_s,\, g_v,\, g_s \odot g_v,\, |g_s - g_v|\,]$ passed through an MLP, optionally augmented with RDKit descriptors, Morgan fingerprints, and an ionic-context feature. All heads below consume this pair representation; the DirectGNN control consumes the identical representation.

5.3 Crystal head and the ideal term

A fusion head predicts the two crystal properties that define the ideal (activity-free) solubility, with physically bounded parametrisations $T_m = 100 + 600\,\sigma(\cdot) \in [100, 700]\,\text{K}$ and $\Delta H_{\text{fus}} = S_H\,\text{softplus}(\cdot) > 0$; an optional entropy-coupled mode instead predicts a Walden-constrained fusion entropy and sets $T_m = \Delta H_{\text{fus}}/\Delta S$. When a group-contribution T_m prior is used it is affine-calibrated on the training set before entering the head. The crystal properties give the ideal term

$$\Phi(T) = \frac{\Delta H_{\text{fus}}}{R}\left(\frac{1}{T} - \frac{1}{T_m}\right), \tag{3}$$

the constant-ΔC_p correction omitted (its magnitude is audited in the SI, §E).

5.4 Three interchangeable activity closures

The activity term $\ln \gamma_2$ is produced by one of three closures, selected by configuration, all differentiable and all consuming the same pair representation.

NRTL. A head predicts the two interaction energies and a non-randomness factor, $\tau_{12}^{\text{ref}}, \beta_{12}, \tau_{21}^{\text{ref}}, \beta_{21}, \alpha \in [0.1, 0.6]$, with a temperature law $\tau_{ij}(T) = \tau_{ij}^{\text{ref}} + \beta_{ij}(1/T - 1/T_{\text{ref}})$; an optional UNIFAC group-contribution prior regularises the τ's. This is the default and the most expressive activity model.

Single-parameter (`gamma_inf`). Collapsing the NRTL closure to a symmetric one-parameter form yields the analytic infinite-dilution law $\ln \gamma_2(x_2, T) = \ln \gamma_2^{\infty}(T)\,(1 - x_2)^2$, with $\ln \gamma_2^{\infty}$ affine in $1/T$. This is the minimal activity model and the setting in which the identifiability analysis is cleanest (§4).

Differentiable COSMO-SAC. A σ-profile head emits, for each molecule, a 51-bin screening-charge-density profile $p(\sigma)$ (area per bin, summing to the cavity area A). A parameter-free COSMO-SAC layer (Lin–Sandler / VT-2005 conventions) maps the pair of profiles to $\ln\gamma_2$ by a damped fixed point for the segment activity coefficient,

$$\Gamma(\sigma_m) = \Big[\textstyle\sum_n p_{\text{norm}}(\sigma_n)\,\Gamma(\sigma_n)\,\exp\big(-\Delta W(\sigma_m,\sigma_n)/RT\big)\Big]^{-1}, \tag{4}$$

with a precomputed exchange-energy matrix ΔW (electrostatic misfit plus a hydrogen-bonding term); the restoring contribution is $\ln\gamma_2^{\text{res}} = (A_2/a_{\text{eff}})\sum_m p_2^{\text{pure}}(\sigma_m)[\ln\Gamma_{\text{mix}}(\sigma_m) - \ln\Gamma_2^{\text{pure}}(\sigma_m)]$, and an optional Staverman–Guggenheim combinatorial term uses the molecular cavity volume and area. The combinatorial size factor $r_i = V_i/r_0$ takes the *true* COSMO cavity volume (V_{cosmo}) so that it shares the cavity basis of the area A_i. This closure — COSMO-SAC evaluated over a network- *predicted* σ-profile, and the intervention that swaps that profile for the measured one — is the object of study in the grounding analysis (§6.1).

5.5 The SLE solver and implicit differentiation

Given $\Phi(T)$ and a closure for $\ln\gamma_2$, the saturated mole fraction solves $\ln x_2 = -\Phi(T) - \ln\gamma_2(x_2, T)$, a fixed point in x_2 because $\ln\gamma_2$ depends on composition. We iterate a damped map

$$x_2^{(k+1)} = \lambda\,\exp\big(-\Phi(T) - \ln\gamma_2(x_2^{(k)}, T)\big) + (1-\lambda)\,x_2^{(k)}, \tag{5}$$

with closure-specific kernels for the NRTL, `gamma_inf`, and COSMO-SAC paths. Rather than back-propagating through the iterations, gradients use the implicit-function theorem at the converged $x_2^\star$,

$$\frac{\partial \ln x_2^\star}{\partial\theta} = -\frac{\partial(\Phi + \ln\gamma_2)/\partial\theta}{1 + x_2^\star\,\eta}, \qquad \eta = \frac{\partial\ln\gamma_2}{\partial x_2}, \tag{6}$$

which is exact at the fixed point, memory-constant in the iteration count, and exposes an explicit stability criterion: the gradient is well conditioned unless $1 + x_2^\star\eta \to 0$. Iteration counts (train 16, eval 30) were set by a convergence audit — a shorter n=8 schedule failed a 5.86-$\ln x_2$ convergence test on strongly non-ideal pairs (SI §A).

5.6 Adaptive correction and the grounding intervention

A bounded, gated "adaptive physics correction" perturbs the physical parameters ($T_m, \Delta H_{\text{fus}}, \tau$) rather than the output, re-solves (5), and blends $\ln x_2^{\text{final}} = \ln x_2^{\text{phys}} + (1-w)\,\text{clip}(\Delta)$; keeping the correction in parameter space preserves the physical decomposition. The *grounding intervention* (a σ-oracle) replaces the predicted σ-profile $\hat{z}$ with the true VT-2005 profile $z^\star$ before the closure, matched by canonical SMILES. It is applied three ways to rule out wiring artifacts: at evaluation only, injected at training time so the rest of the model co-adapts, and as a channel-swap control; an analogous override substitutes true crystal targets. The activity target used for the closure/insufficiency decomposition (§6.2) is evaluated at infinite dilution, $\ln\gamma_2^\infty = \lim_{x_2\to 0}\ln\gamma_2$, at 298 K.

5.7 Training: three-phase curriculum and single-component grounding

Optimisation follows a fixed three-phase curriculum: (P1) auxiliary-property pretraining on crystal, Hansen, and $\ln\gamma_2^\infty$ targets with no SLE loss; (P2) full SLE training, with the adaptive correction unfrozen partway and an optional crystal→pair weight ramp; (P3) low-learning-rate stabilisation with monotonicity and van't Hoff regularisers. External single-component data enter through *grounding streams*: each stream is a sidecar of *self-solvent* rows (solvent = solute, no solubility label, only the target factor's mask active), interleaved into P1/P2 at a configurable step budget, with a mandatory scaffold-leak guard excluding any pool molecule whose scaffold is in validation or test. The reference streams ground the crystal factor (melting-point pool) and the activity factor (σ-profile database); this single-component-grounding pattern is the mechanism by which an otherwise non-identifiable factor (§4) is pinned by abundant external data a black-box predictor cannot use.

5.8 Objective

The training loss is a weighted sum with per-phase weights. Only a small set of terms is load-bearing for the claims in this paper: a bin-weighted Huber on $\ln x_2$; auxiliary losses on the σ-profile (an earth-mover distance), on crystal $T_m/\Delta H_{\mathrm{fus}}$, on $\ln \gamma_2^\infty$, and on Hansen parameters; and structural regularisers for temperature monotonicity and the van't Hoff slope. The remaining terms (channel orthogonality, contrastive, mixture-of-experts balance, and various priors) are auxiliary; the complete taxonomy and weights are tabulated in the SI (§A).

5.9 The DirectGNN control

DirectGNN shares the encoder, interaction, readout, and pair representation, adds a thermometer temperature encoding, and maps directly to $\ln x_2$ with a bin-weighted Huber loss and no heads, closure, or solver. Because it shares everything but the thermodynamic bottleneck, the TGNNSolv$-$DirectGNN gap is attributable to the bottleneck rather than to representation, and DirectGNN is the reference for every "physics tax" comparison in §6.

5.10 Reproducibility

All controlled comparisons are run at three seeds and reported as mean±sd on a cross-arm intersection lock (rows supervised and finite in every arm). Solver iteration counts, COSMO-SAC constants, the exact feature list, per-phase loss weights, and the compute ledger (which results are full-corpus GPU runs versus CPU-reproducible proxies) are given in the SI (§A, §F); the named analysis scripts regenerate every figure and table from committed artifacts.

6 Results

6.1 The grounding paradox

Feeding true σ-profiles into the closure degrades it rather than improving it, and it does so in both the activity and the solubility target. On the activity target ($\ln \gamma_2^\infty$, n=60 pairs with both true profiles present), a flexible regressor of $\ln \gamma_2^\infty$ on the true profiles—our irreducible-variance (B_{insuff}) estimator—attains MSE ≈ 0.40, whereas the fixed closure on the *same* true profiles attains MSE 1.47–1.80 depending on convention (§6.2): about 4$\times$ above the irreducible-variance floor, and the true-input closure under-predicts $\ln \gamma_2^\infty$ with a systematic downward bias (Fig. 3). In the solubility target the effect is large enough to erase any positive R^2 (Fig. 2, three seeds, mean±sd, n=5608): the model predicts solubility *better with its own learned σ-profile* (grounded MAE 1.85 ± 0.05, $R^2 = 0.33 \pm 0.05$) than when *given the true measured one* (the σ-oracle, MAE 2.25 ± 0.02, $R^2 = -0.03 \pm 0.04$, no better than predicting the mean; one of three seeds is marginally positive). The robust form of the effect is the MAE gap (2.25 vs 1.85); the near-zero oracle R^2 shows the true input carries

Figure 2: Controlled comparison on the scaffold split (3 seeds, mean±sd, n=5608; metrics are the 3-seed mean of the cross-arm intersection lock — rows supervised and finite in every arm — from `run_e5_comparison.py`). DirectGNN shares the encoder but has no thermodynamic closure; the physics closures trail it (the "physics tax"). Replacing the model's learned σ-profile with the *true* VT-2005 profile (the σ-oracle, salmon) gives the *worst* arm, collapsing R^2 to ≈ 0 (-0.03 ± 0.04): by MAE the model predicts solubility distinctly better from its invented σ (1.85) than from the measured one (2.25). Per-arm exact values are transcribed in the SI.

no usable solubility signal through this closure. Adding the physically *more* complete Staverman–Guggenheim combinatorial term also increases error (activity MSE 1.47 → 1.80; solubility MAE 1.85 → 1.88), which is what a misspecified map does and what missing information would not. We confirm the paradox three independent ways: the evaluation-time oracle above, injecting the true profile *at training* (single-seed MAE 1.98, still worse than the learned-σ 1.80), and a channel-swap control (MAE 2.08), ruling out a train/test wiring artifact.

6.2 Measuring the ceiling: the closure dominates

We apply the decomposition of §4.2 to the n=60 matchable pairs (41 solutes, 17 solvents; the two formamide solvents that the deployed canonical-SMILES matcher drops on a tautomer mismatch are excluded), with $z^\star$ the concatenated true solute/solvent σ-profiles (102 dimensions) and m the experimental $\ln \gamma_2^\infty$ (298 K, IDAC). Table 1 reports the bounds.

The load-bearing separation (convention-independent, Tier 1). Because $B_{\text{insuff}} = \mathbb{E}[\text{Var}(m \mid z^\star)]$ depends only on $(m, z^\star)$, its upper bound is the same in every closure convention; subtracting it from the total therefore lower-bounds B_{closure} even for the model's *deployed* residual-only convention: $B_{\text{closure}} \geq \text{MSE}_{\text{res}} - B^{\text{up}}_{\text{insuff}} \approx 1.47 - 0.62 = 0.85$ (and 0.91 against the RF bound), well above $B_{\text{insuff}} \leq 0.62$. This is the claim we lean on, and it needs no smoothness assumption beyond the validity of the B_{insuff} upper bound. Under the full COSMO-SAC convention the separation *sharpens* to an assumption-free constant-offset bound: a constant decoder offset alone accounts for $B_{\text{closure}} \geq 0.72$ (Fig. 4), exceeding every B_{insuff} upper bound (≤ 0.62), and subtracting the insufficiency bound gives $B_{\text{closure}} \geq 1.24 > 0.62$ ($\sim 2\times$). That constant-offset bound is convention-specific

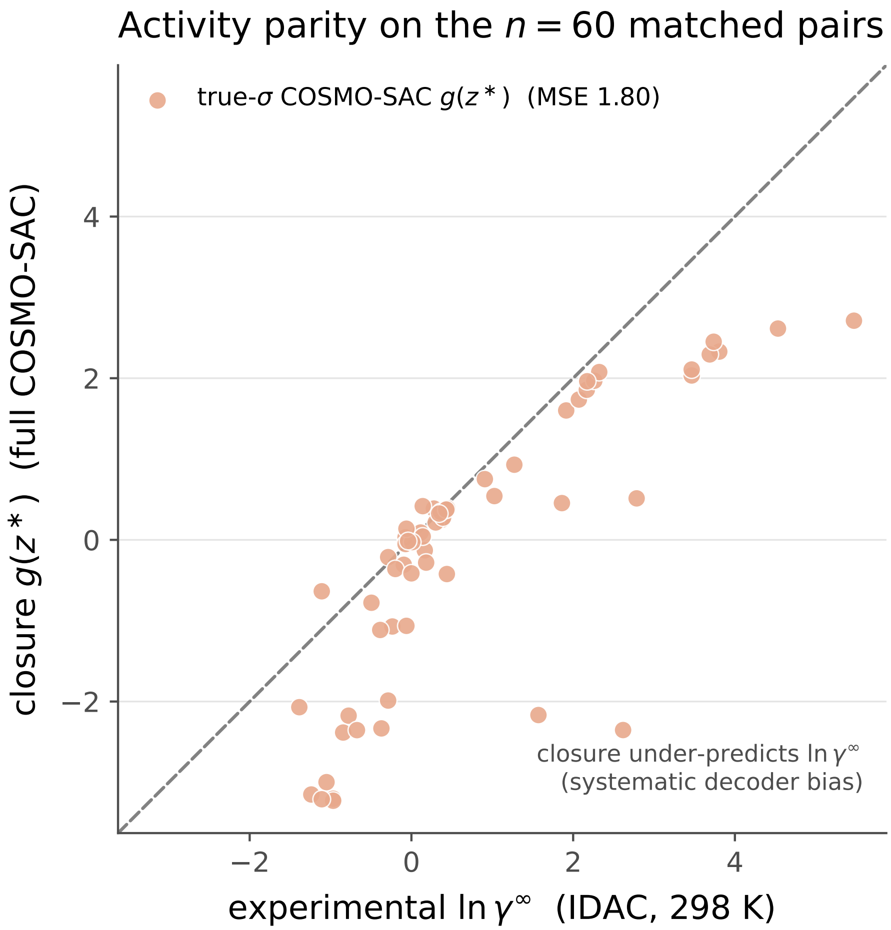

Figure 3: Activity-coefficient parity on the n=60 matched pairs. The true-σ COSMO-SAC closure $g(z^\star)$ (salmon) falls systematically below the diagonal at larger $\ln\gamma_2^\infty$, under-predicting the activity coefficient (MSE 1.80, full convention). This systematic downward bias is the visible form of the decoder misspecification B_{closure}; for reference a flexible regressor of $\ln\gamma_2^\infty$ on the same true profiles reaches the irreducible-variance floor (MSE $\approx$ 0.40, our B_{insuff} estimator; §6.2), so the closure's excess over that floor is B_{closure}.

and *inverts* under the deployed residual-only default ($0.43 < 0.62$; hedge 1), which is exactly why we headline the convention-independent bound above rather than the constant offset. Label noise is unestimable in this replicate-free set and assumed zero; the bounds we rely on are invariant to zero-mean noise, so the ceiling is not label noise either.

Convention-robust corroboration (Tier 2). The genuinely convention-robust fact is that adding the physically *more* complete Staverman–Guggenheim combinatorial term *increases* error (activity MSE 1.47 → 1.80)—what a misspecified map does, not what missing information would. This is robust to the size-factor's volume basis: recomputing the combinatorial term on the true COSMO cavity volumes (VT-2005 V_{cosmo}) rather than an RDKit molar volume leaves it essentially unchanged (Jensen 0.714 → 0.717, MSE 1.793 → 1.800), because the term depends on molecular-size *ratios* in which a near-uniform volume rescaling cancels. A second observation is consistent but is *not* independent of the decomposition: a flexible regressor of m on $z^\star$ fits the activity target $\sim$4×

Table 1: Closure–insufficiency decomposition on the true-input activity path. Bounds, not a point split (Lemma 3). Convention "full" = COSMO-SAC with the combinatorial term (size factor $r_i = V_i/r_0$ on the true COSMO cavity volume, VT-2005); "res" = residual-only (the deployed default). B_{insuff} upper bounds are convention-independent estimates of $\mathbb{E}[\mathrm{Var}(m \mid z^{\star})]$; "kNN" is a direct high-dimensional estimate (biased upward), the others are law-of-total-variance / out-of-fold upper bounds. The *load-bearing* separation is the convention-independent $B_{\text{closure}} \geq \text{MSE} - B_{\text{insuff}}^{\text{up}}$ row, which exceeds B_{insuff} under *both* conventions (1.24, 0.91 > 0.62); the assumption-free constant-offset (Jensen) bound separates only under "full" (0.72) and inverts under the deployed "res" (0.43 < 0.62).

Quantity	full	r
$\text{MSE}_{\text{total}} = B_{\text{closure}} + B_{\text{insuff}}$ (exact)	1.80	1.4
B_{closure} lower via $\text{MSE}{-}B_{\text{insuff}}^{\text{up}}$ (RF; B_{insuff} *conv.-indep., load-bearing*)	1.24	0.9
B_{closure} lower bound (Jensen, offset^2; assumption-free; *full-only*)	0.72	0.4
B_{insuff} upper (bin g; law of total variance)	0.53	0.6
B_{insuff} upper (RF / Ridge out-of-fold)	0.56/0.62	sam
B_{insuff} (kNN-in-$z^{\star}$, convention-indep., biased up)	0.40	0.4
label noise σ_η^2 (IDAC, unestimable here)	assumed 0 (no replicates in matched se	

better than the fixed true-input closure—but that regressor is exactly our convention-independent B_{insuff} estimator (≈ 0.4), so "4× better" restates that the closure sits far above the irreducible-variance floor (that B_{closure} is large) rather than corroborating it by an independent route. We therefore lean on the dimension-robust random-forest / ridge out-of-fold and binning bounds and on the combinatorial effect, which agree that B_{closure} exceeds B_{insuff}.

Four hedges we disclose. The adversarial audit (four independent recomputations) could not invert the verdict but sharpened its scope:

1. *Convention.* The deployed default is residual-only. Under it the *constant-offset* (Jensen) lower bound inverts ($B_{\text{closure}} \geq 0.43 < B_{\text{insuff}} \leq 0.62$), so that particular assumption-free bound does not separate, and the $\sim 2\times$ full-convention margin (the sharpened $\text{MSE} - B_{\text{insuff}}$ bound, 1.24 vs 0.62) is a full-COSMO-SAC-only result. What separates under the deployed convention is the convention-independent bound $B_{\text{closure}} \geq \text{MSE}_{\text{res}} - B_{\text{insuff}}^{\text{up}} \approx 0.91 > 0.62$, on which we headline, together with the combinatorial Tier 2 fact—so the headline does not rest on the arm-specific constant-offset arithmetic.

2. *Confound.* The closure error is predominantly a *between-solvent* systematic bias (~64% of total error); within the largest single-solvent stratum (pyridine, n=20) the closure is nearly correct (stratum MSE 0.07). Because solvent σ is part of $z^{\star}$, these offsets are still decoder error and give an independent solvent-conditional lower bound $B_{\text{closure}} \geq 0.89$ (full)/0.56 (res).

3. *Confidence.* Honest confidence on the full-convention constant-offset separation is the solvent-clustered bootstrap $P(B_{\text{closure}} > B_{\text{insuff}}) \approx 0.85$ (0.83 against the random-forest upper bound, 0.89 against the law-of-total-variance bound); we report the solvent-clustered scheme rather than the less conservative i.i.d.-pair one because solvent dependence dominates. The convention-independent margin (0.91 vs 0.62) is reported as a point estimate and is the direction we

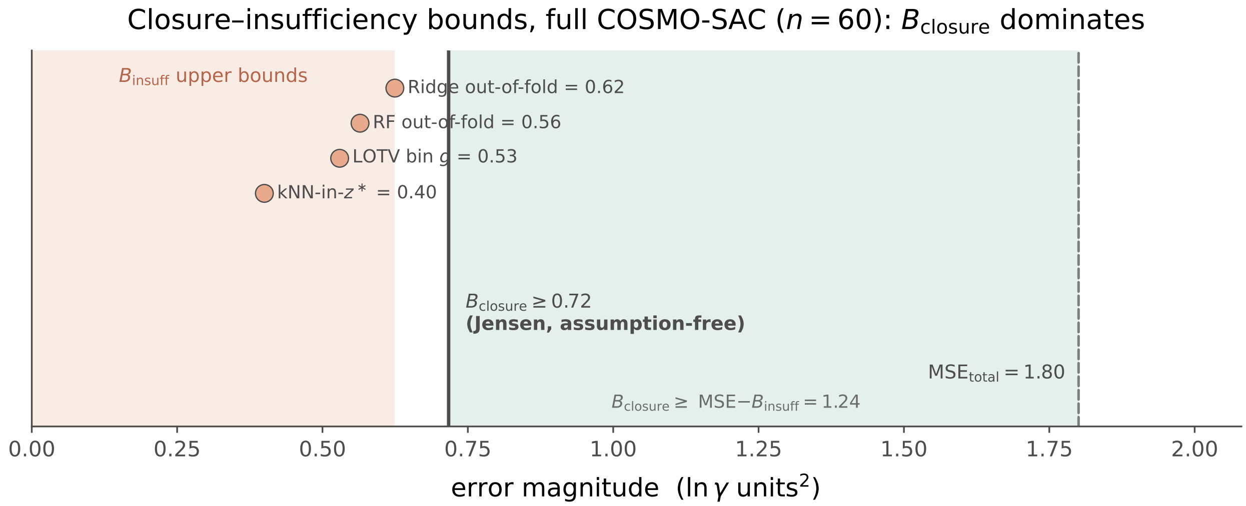

Figure 4: One-sided bounds on the closure–insufficiency split (full COSMO-SAC convention). The assumption-free Jensen lower bound on the decoder misspecification, $B_{\text{closure}} \geq 0.72$, exceeds every valid upper bound on the input insufficiency $B_{\text{insuff}} \leq 0.62$ (four independent estimators cluster in the salmon region); the white gap is a separation that holds with no smoothness assumption. The teal region is the closure share of the total error (MSE = 1.80). This assumption-free separation is convention-specific; under the deployed residual-only default it rests instead on the convention-independent $B_{\text{closure}} \geq \text{MSE} - B^{\text{up}}_{\text{insuff}} \approx 0.9 > 0.6$ bound (§6.2).

headline. The decomposition is checkpoint/seed-independent (the closure has zero learnable parameters).

4. *Selection/scope.* The 60 matchable pairs are the low-to-moderate γ corner (matched mean $\ln \gamma_2^\infty = 0.75$ vs 3.56 for VT-2005-unmatchable pairs, $t = -7.44$; the strongly non-ideal amide-solvent regime is untested because VT-2005 lacks those profiles). This narrows scope but is *conservative*, since the untested regime is a harder closure test. The target is $\ln \gamma_2^\infty$ at 298 K, so the numeric ceiling is not claimed to transfer unchanged to the finite-composition solubility regime.

6.3 Closure fidelity sets the true-input penalty

To isolate the mechanism behind the grounding paradox, we constructed a controlled synthetic experiment in which the only quantity that varies is the fidelity of the closure, with input quality, sample size, and estimator held fixed. A teacher draws a latent variable $z^\star$ and generates a target through a known map; a candidate closure then reconstructs the target from $z^\star$ under a tunable misspecification that deforms the true map by a controlled amount. We summarize the deformation by a scalar fidelity F, where $F = 1$ recovers the exact map and F decreases as the closure is pushed further from the teacher. The measurement of interest is the true-input penalty $\Delta R^2 = R^2_{\text{true-input}} - R^2_{\text{learned-head}}$, the change in fit obtained by feeding the genuine latent $z^\star$ through the closure rather than fitting a free head to the target. A negative value means that supplying the correct physical input actively hurts, which is the signature we observe empirically when the VT-2005 σ-profiles are pushed through the COSMO-SAC closure. As F falls across its physical range, the penalty grows monotonically and without exception: $F = 1.00$ gives $\Delta R^2 = 0$; $F = 0.76$ gives -0.096; $F = 0.38$ gives -0.247; and pushing the map past physical fidelity as an extreme

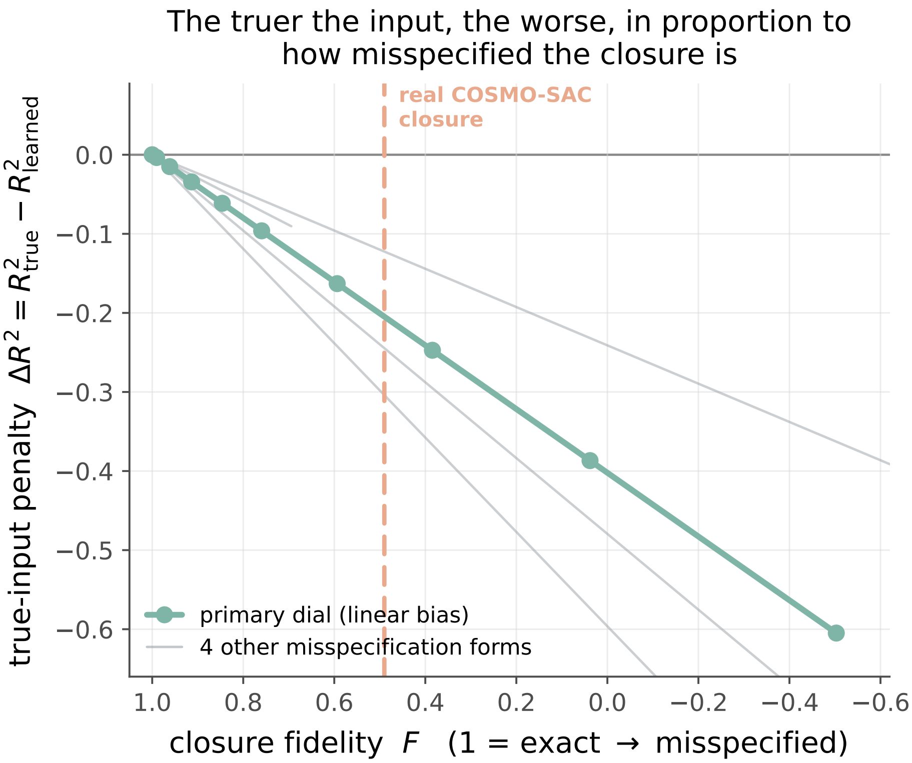

Figure 5: Synthetic closure-fidelity dial. As the closure is made more misspecified (fidelity F decreasing from 1), the true-input penalty $\Delta R^2 = R^2_{\text{true}} - R^2_{\text{learned}}$ grows monotonically, and the same ordering holds across five misspecification functional forms (grey). The real COSMO-SAC closure sits at $F \approx 0.49$ (salmon), well inside the misspecified regime.

stress test ($F = -0.50$, a formal over-rotation of the true map) gives -0.605 (Fig 5). The same monotone ordering holds across every misspecification functional form we tested, spanning linear bias, quadratic, cubic, absolute-value, sinusoidal, and cross-term deformations, so the effect is a property of the fidelity magnitude and not an artifact of one particular way of breaking the map.

This construction separates the two error channels that the physics path confounds. When the closure is exact, the true input is optimal and there is nothing to gain by fitting a free head, so $\Delta R^2 = 0$; the penalty appears only as the closure becomes misspecified, and its size is set by that misspecification alone rather than by any deficiency in the input. Placing the real system on the same dial only illustratively—the mapping from a real closure to a scalar fidelity F is a synthetic construction, not an independently calibrated measurement—the COSMO-SAC closure over predicted σ-profiles falls well inside the misspecified regime (F below ~ 0.5), where the dial predicts a substantial negative true-input penalty of the same sign as the empirical grounding paradox, whose solubility ΔR^2 between the σ-oracle and learned-σ arms is about -0.36 (which on the dial's own monotone curve corresponds to $F \lesssim 0.4$; we do not claim a precise fidelity value). This locates the empirical grounding paradox on a mechanistic axis: the loss from feeding true physical inputs is what one expects from a closure of this fidelity, consistent with B_{closure} dominating B_{insuff} in the decomposition, and it identifies closure fidelity, not input quality, as the quantity that governs whether grounding helps or harms.

Table 2: Fisher audit of the SLE design at $\Delta T = 40\,\mathrm{K}$ (synthetic, closure-independent). Solubility alone leaves the crystal parameter near-unidentified; external single-component labels close the deficiency.

Supervision regime	cond(I)	CRLB sd(ΔH_{fus}) [J/mol]
Solubility only	2.3×10^5	19,895
+ T_m label	5.4×10^3	3,151
+ $T_m + \Delta H_{\mathrm{fus}}$	1.6×10^3	1,689

6.4 The ceiling is not fixable downstream (Gate B)

If the closure binding were merely a bad point estimate, a learned output-space correction on top of the physics prediction should recover accuracy. It does not. An additive bounded residual with a learned confidence gate ($\hat{y} = \hat{y}_{\mathrm{phys}} + (1-c)\,r$, with $y = \ln x_2$) leaves the valve effectively shut (applied residual ≈ 0, $R^2 = 0.236$). Forcing the gate open removes the systematic *bias* (from +0.45 to +0.06) but does *not* recover *accuracy* (R^2 $0.283 \to 0.274$). This is the classical calibration-versus-accuracy separation: when the upstream map is misspecified, a monotone or affine post-hoc map can fix reliability but not resolution. Recalibration buys calibration without buying accuracy, which is what we expect if the closure, rather than the estimate, is the binding constraint.

6.5 Identifiability and compensation

Lemma 1 asserts that the crystal/activity split is not recoverable from solubility data alone. Two diagnostics make this concrete. The first is a numerical Fisher audit of the SLE design, computed on synthetic data and independent of any particular activity closure. We evaluate the Fisher information matrix I of the observation model at a reference operating point and read off the conditioning cond(I) and the Cramér–Rao lower bound on the standard deviation of ΔH_{fus} under three supervision regimes. Table 2 reports the result at a temperature window of $\Delta T = 40$ K. With solubility measurements only, the design is severely ill-conditioned and the CRLB on ΔH_{fus} is roughly 20,000 J/mol, comparable to the quantity being estimated. Adding a direct T_m label drops the conditioning by nearly two orders of magnitude and cuts the bound to about 3,200 J/mol; supplying T_m and ΔH_{fus} together brings it to about 1,700 J/mol. The ill-conditioning is a property of the design rather than of noise level, and it is closed only by external single-component supervision, which is the mechanism the aux-stream pattern supplies.

Widening or narrowing the temperature window does not rescue the solubility-only case. A ΔT sweep moves the solubility-only conditioning from 1.7×10^3 at a wide window to 2.3×10^5 at 40 K and to 1.9×10^{16} at 10 K, so multi-temperature data alone does not resolve the split; the two parameters remain confounded along the ideal-term direction. Restricting the activity slope rather than adding labels helps, but only partially: it cuts sd(ΔH_{fus}) to 8,251 J/mol, still well above the label-supervised regimes. A second, model-side diagnostic must be read with more care, and we keep it out of the load-bearing argument. On the supervised test set, with a group-contribution (Joback) crystal reference, the crystal and activity residuals are strongly anti-correlated, $\mathrm{corr}(\delta\Phi, \delta \ln \gamma_2) = -0.962$. This is an *accounting artifact*, not evidence of a learned split: the two residuals sum to the (negative) $\ln x_2$ fit residual, so their correlation tends to -1 as the fit improves, and a permutation null that destroys any learned coupling still reproduces most of it (-0.90 to -0.93); the model's own factors barely co-vary ($\mathrm{corr}(\hat{\Phi}, \ln \gamma_2) = -0.19$). The large apparent crystal offset ($\delta\Phi \approx -7.1$) is likewise not a usable mis-grounding metric — it is dominated by reference error, since the Joback reference is physically implausible on these drug-like polycyclics (a substantial fraction of predicted T_m exceeds 1000 K), and capping it at a physical bound collapses the offset to ≈ -1.5. We therefore rest the non-identifiability claim on the reference-free Fisher/CRLB audit above, and relegate the compensation figures to the SI (§E) as a cautionary diagnostic rather than a decomposition-quality measurement.

6.6 Grounding the crystal factor: a dose-limited null

A natural test of whether the ceiling is inputs rather than closure is to ground a *different* factor—the crystal term—on abundant external data. We trained the identical model without and with an external single-component crystal-property pool of roughly 15,000 melting points, added through the auxiliary-stream mechanism after excluding any pool solute whose Bemis–Murcko scaffold appears in the validation or test split. The pool did not improve prediction of the crystal properties it supplies (melting-point MAE 45.0 $\rightarrow$ 46.6 K, effectively unchanged, if anything marginally worse), and downstream solubility did not improve either ($\ln x_2$ MAE 1.81 $\rightarrow$ 1.88; the two nominally favorable shifts—$\ln x_2$ R^2 0.320 $\rightarrow$ 0.355 and crystal offset $|\delta\Phi|$ 4.99 $\rightarrow$ 4.49—are slight and do not amount to a gain). We flag this as *dose-limited and inconclusive rather than a clean null.* In this run the crystal auxiliary stream was homeopathically dosed—a small fraction of optimizer steps at low loss weight, so the model saw the external pool less than one full pass at the selected checkpoint—and a stream that did not even lower the MAE of the property it directly supervises cannot license a strong conclusion. What it shows is that *this dilute intervention* does not move the ceiling, not that external crystal labels cannot. We therefore do *not* read it through the activity-path $B_{\text{closure}}/B_{\text{insuff}}$ decomposition: the crystal "closure" is the near-exact ideal-solubility term $\Phi = (\Delta H_{\text{fus}}/R)(1/T - 1/T_m)$, not a misspecified map, so the flat T_m recovery is better read as the crystal/activity split's structural non-identifiability (Lemma 1) and the encoder's transfer limitation (§6.9) than as closure dominance. A properly dosed crystal-grounding test is left to future work.

6.7 Grounding does not pay off in the low-data regime either

The strongest a-priori case for a physics prior is data scarcity: with few training molecules, an inductive bias toward the correct physical factorization should beat a data-hungry black box even when the two tie at full data. We test this directly. Subsampling the training set *by solute* at fractions 0.05–1.0 (the regime that governs scaffold transfer), we train the physics-grounded model and DirectGNN on the identical subsample and evaluate both on the fixed scaffold test (Table 3; seed 42, full-corpus run). The physics arm is *worse at every fraction*, and—against the hypothesis—the gap is *smallest* at the least data (ΔMAE $= +0.08$ at 5%) and *grows* with more ($+1.0$ at full); its R^2 is negative throughout. Even in the regime built to favor it, grounding does not buy accuracy; if anything, more data lets the black box pull further ahead. The one metric that inverts is top-1 solvent ranking, higher for the physics arm at 5% (0.42 vs 0.23)—consistent with the ranking-versus-accuracy split of §6.9—but that decision-quality signal does not translate into lower error. This is the last regime in which a physics prior should plausibly have helped; that it does not closes the accuracy case and leaves the structural account of §4 and §6.2—a non-identifiable split fed through a misspecified closure—as the operative explanation.

6.8 Where physics helps and hurts

Across the full test set the two arms are separated by little more than label noise. On the 3-seed scaffold split the DirectGNN control reaches an $\ln x_2$ MAE of 1.70 ± 0.03, while the grounded physics

Table 3: Data-efficiency sweep (train subsampled by solute, fixed scaffold test, seed 42). The physics-grounded arm trails DirectGNN at every fraction and the gap widens with data. ΔMAE = MAE(physics) − MAE(direct) > 0 means physics is worse. Absolute errors here exceed the headline runs (a lighter per-fraction budget); the valid signal is the within-budget physics−direct gap.

train fraction	MAE		ΔMAE	R^2	
(by solute)	physics	direct	(phys−dir)	physics	direct
0.05	2.28	2.20	+0.08	−0.09	+0.07
0.10	2.44	2.07	+0.37	−0.25	+0.15
0.25	2.93	1.81	+1.12	−0.75	+0.38
0.50	3.12	1.98	+1.14	−0.98	+0.22
1.00	2.98	1.97	+1.01	−0.79	+0.22

arm (learned-σ COSMO-SAC closure) reaches 1.85 ± 0.05, for a global $\Delta\text{MAE} = \text{MAE(physics)} - \text{MAE(control)} = +0.14$ (positive meaning the closure adds cost; the per-seed values span +0.05 to +0.25, so this point estimate is itself only a 3-seed mean, not a tight one). At this granularity the physics bottleneck is a small, near-noise tax on accuracy. The interesting structure appears only once the error is resolved along chemical axes, where the average tax turns out to be an average over regimes that disagree in sign.

The solvent axis is the sharp one. Binning test pairs by solvent class and recomputing ΔMAE per seed shows the cost concentrated in solvents that hydrogen-bond strongly and directionally, precisely the regime in which the COSMO-SAC closure is least faithful and $|\ln\gamma_2|$ is largest. Water carries the heaviest penalty, and the effect decreases monotonically as the donor strength of the class softens. One class inverts the sign: amide solvents, which are aprotic dipolar hydrogen-bond acceptors and therefore closer to the closure's well-specified regime, are the stratum where the physics arm most consistently helps (-0.22 ± 0.11 over seeds). We flag this as exploratory rather than a confirmed win: across a family of nine simultaneous solvent-class comparisons the amide interval is an uncorrected per-class pair bootstrap, and the three-seed spread alone does not exclude zero; the halogenated class shows a nominal improvement of similar magnitude but is not seed-robust. The robust reading is the *direction* — the physics tax shrinks and can turn negative as solvent hydrogen-bond donation weakens toward the acceptor-dominated regime the closure was built for — not the significance of any single class. This solvent-resolved pattern is the chemistry-resolved shadow of the same synthetic dial we used to vary closure fidelity in controlled experiments.

The solute axis tells a compatible but weaker story in prose. Oxygenated solutes (+0.32), polyaromatics (+0.27), and heterocycles (+0.17) are where the closure costs the most, while halogenated-aromatic solutes are roughly neutral. An earlier hypothesis, that the physics path should hurt most on charged and high-$|\ln\gamma_2|$ solutes such as salts, did not replicate once the split was corrected: the charged/salt stratum is neutral at -0.04 ± 0.19. The clean penalty falls instead on ordinary neutral organics, not on the ionic tail we had expected to be hardest.

The most suggestive pattern is a novelty inversion that runs opposite to the usual coverage intuition. Binning solutes into three pre-specified nearest-train Tanimoto-similarity strata, the physics arm is neutral on the most novel chemistry (≤ 0.3: -0.04 ± 0.18) and hurts most on the most familiar chemistry (> 0.8: $+0.51 \pm 0.09$). We flag this as exploratory, not a confirmed effect: the ± 0.09 is the spread across three training seeds on the fixed test split, not a within-bin sampling interval over which solutes fall in the stratum, and the Tanimoto binning is one of several strata

Table 4: Per-solvent-class physics tax $\Delta\text{MAE} = \text{MAE(physics)} - \text{MAE(control)}$ on the scaffold split (mean ± sd over 3 seeds; positive means the closure adds cost). The intervals are uncorrected per-class pair bootstraps across a family of nine simultaneous comparisons; the two negative (physics-helps) entries are exploratory, not multiplicity-controlled.

Solvent class	ΔMAE (mean ± sd)
Water	$+0.52 \pm 0.35$
Carboxylic acid	$+0.39 \pm 0.16$
Sulfoxide	$+0.26 \pm 0.24$
Nitrile	$+0.21 \pm 0.19$
Alcohol	$+0.15 \pm 0.14$
Aromatic	$+0.03 \pm 0.63$
Hydrocarbon	$+0.02 \pm 0.50$
Halogenated	-0.29 ± 0.36
Amide	-0.22 ± 0.11

families we examine (solvent class, solute class, similarity) without multiplicity control. Read as a direction rather than a tested quantity, it suggests the ceiling on the physics path is not coverage or extrapolation but the closure's systematic misfit—masked on novel solutes by the large errors both arms already incur there, and exposed on easy, well-covered solutes where the direct control is accurate enough for a structural bias in the closure to become the dominant residual—consistent with the same closure-misspecification reading as the rest of the paper. A multiplicity-controlled, within-bin-bootstrapped test is left to future work. Read together, the maps say the physics decomposition pays where the closure is misspecified relative to the solvent chemistry, and it can only turn into a win in the narrow acceptor-dominated regime the closure was built to describe.

6.9 Supporting diagnostics

The interpretive claims of the preceding sections rest on a set of controlled diagnostics, each run under a named protocol. We report them here with their exact numbers, including the results that run against the physics-informed hypothesis.

Aleatoric noise floor. Repeated solubility measurements of the same solute in the same solvent disagree across laboratories, and that disagreement sets a floor below which no model can be expected to fit. Measuring inter-source spread on [8], the mean absolute disagreement is approximately 0.15 $\ln x_2$ units for pairs backed by two sources ($n = 3948$ groups), rising to 0.31 for groups backed by three or more sources ($n = 349$); the corresponding medians are smaller still (0.03 and 0.09). The scaffold-split MAE we report (≈ 1.7 to 1.9) sits well above this band, so label noise does not account for the accuracy ceiling. The floor is small; the ceiling is not a data-quality artifact.

The encoder is transfer-limited, not expressivity-limited. To separate what the frozen graph embedding can represent from what it transfers out of distribution, we fit a linear probe from the encoder representation onto RDKit descriptors and compare in-distribution and held-out fit. The median train R^2 is 0.758 against a test R^2 of 0.448 ($n = 700$ solutes in each partition), a gap of 0.31. Individual descriptors split sharply: FractionCSP3 transfers well (test $R^2 = 0.96$) while HeavyAtomCount does not (0.23). The embedding is expressive enough in sample; it is the generalization of that expressivity across scaffolds that fails. Swapping in a new graph encoder is therefore unlikely to lift out-of-distribution accuracy, and the useful levers are pretraining and coverage rather than architecture.

Temperature extrapolation. A structured van't Hoff activity class encodes the correct temperature dependence of $\ln \gamma_2$ by construction, and this shows up when the split withholds temperature rather than structure. Training on measurements at $T \leq 310$ K and testing at $T \geq 330$ K ($n_{\text{test}} = 3343$), the structured class attains a high-temperature MAE of 0.368 ($R^2 = 0.887$) against a random forest over Morgan fingerprints plus temperature at 1.290 ($R^2 = 0.658$), consistent with the temperature-aware $\ln \gamma_2^\infty$ modelling of [12]. This is a property of the structured hypothesis class, not a performance claim we make for the trained model; the point is that the functional form carries temperature extrapolation that a descriptor-plus-temperature regressor does not.

Ranking and calibration. Absolute error and decision quality come apart. Despite mediocre MAE, the physics path orders candidate solvents usefully: Spearman correlation 0.511, top-1 accuracy 0.477, Kendall τ 0.436, and nDCG@3 0.732 across $n = 826$ solute groups. Uncertainty, by contrast, is badly miscalibrated in the native model, with a nominal-90% interval covering only PICP@90 = 0.127 of held-out points. Split-conformal wrapping repairs this: at a nominal level of 0.90 the empirical coverage is 0.918 on $n = 5826$ held-out points (the full scaffold test set, larger than the $n = 5608$ cross-arm intersection used for the controlled comparison of Fig. 2). The ranking signal and the post-hoc calibrated intervals are the practically usable outputs, even where the point predictions are not competitive.

Data efficiency is out of scope. A common motivation for a physics prior is that it should pay off when data are scarce. Testing this properly requires a full data-efficiency sweep — subsampling the training set by solute across fractions 0.05 to 1.0 and retraining both arms at each fraction — which is beyond the compute budget of this study; we provide the runner but do not report the curve here. We therefore make no low-data claim. The interpretive value of the decomposition is an attribution affordance, not a demonstrated sample-efficiency advantage, and whether a physics prior buys data efficiency on this corpus is left open.

7 Discussion

The result is a cautionary measurement for physics-informed modeling: grounding a model on truer physical inputs is only as good as the closure that consumes them, and when that closure is a fixed, mildly misspecified map, better inputs surface its bias rather than help. The external latent does not make the physics path win. What it buys is *attribution*: on the infinite-dilution, low-γ subset we can measure, the ceiling sits in the decoder (B_{closure}), with the inputs (B_{insuff}) ruled out as its source and label noise unestimable but assumed zero (and unable to manufacture the constant offset the bounds rest on). We do not claim this numeric attribution transfers unchanged to the finite-composition, high-γ regime the solubility paradox itself spans; that the solubility paradox shares the same closure-misspecification cause is a conjecture supported by in-regime signs—the more-physics-worse effect on the full n=5608 set and the chemistry map—not a measured fact. For practice this relocates the lever. Within this closure family, effort spent on richer inputs or a better encoder is mis-targeted; the payoff is in the closure itself, for example a corrected or learned residual on the map rather than on its output (which §6.4 shows is inert).

The accuracy case is closed, exhaustively. That the physics prior does not improve accuracy is not a single observation but a pattern with no exception we could find: the grounded model trails the matched black box on the full corpus, the structured temperature advantage is a property of the van't Hoff hypothesis class rather than of the trained model, grounding the crystal factor on external labels does not move the ceiling (§6.6), and—in the case built to favour a physics prior—it loses at *every* training-data fraction, including the low-data regime where an inductive bias should help most (§6.7). Read together these are not scattered nulls but evidence that the ceiling is *structural*: the crystal/activity split is non-identifiable from solubility alone (Lemma 1), and the activity closure is misspecified where we can measure it (§6.2), so no amount of better inputs, more data, or auxiliary grounding within this family recovers an accuracy the factorization cannot express. The contribution is accordingly not an accuracy result but a method—the externally grounded decomposition—for locating such a ceiling, and a proved account of why it sits where it does.

Limitations. (i) B_{insuff} is bounded, never point-identified: there are no true replicate measurements at fixed $z^{\star}$, so the insufficiency estimates lean on a smoothness assumption. The constant-offset (Jensen) bound is smoothness-free but assumes m and g share the $\ln \gamma_2^{\infty}$ (mole-fraction, Lewis–Randall) reference state; we check this two ways—the closure is nearly exact within the largest single-solvent stratum (pyridine, MSE 0.07), which a uniform reference-state offset could not produce, and an independent recomputation reproduces the closure output—so the mean offset is genuine decoder bias, not a units artifact. We headline the convention-independent bound (§6.2) as the load-bearing claim. (ii) The assumption-free constant-offset margin is convention-specific (full COSMO-SAC) and *inverts* under the deployed residual-only default; there the separation rests on the convention-independent bound $B_{\text{closure}} \geq \text{MSE} - B_{\text{insuff}}^{\text{up}}$ and the more-physics-worse effect (which we verified is robust to the molecular-volume basis). (iii) The measurement is at infinite dilution and 298 K on the VT-2005-matchable, low-γ subset; the non-ideal amide-solvent regime is untested. (iv) Absolute scaffold $\ln x_2$ accuracy is encoder-bound and near the aleatoric label-noise floor; this paper is not an accuracy claim.

8 Conclusion

Feeding a physics-informed solubility model its true physical inputs made it worse. What the paper contributes is a way to measure that failure: with an externally grounded latent, the true-input excess risk splits into a decoder-misspecification term and an input-insufficiency term, and on the infinite-dilution, low-γ subset we can measure, the ceiling turns out to be the closure rather than the inputs (label noise unestimable and assumed zero). The practical implication is to fix the map itself, since within this closure truer inputs surface its bias and an output-space correction is inert.

Data and code availability

All code and the intermediate result artifacts needed to regenerate every figure and table are available at `[repository URL]` (commit `[hash]`), released under `[license]`. The solubility corpus is BigSolDB 2.0 [8]; the true σ-profiles are the VT-2005 database, ingested to `results/sigma_profile_artifact/sig` the experimental infinite-dilution activity coefficients are compiled from ThermoML (`notebooks/data/raw/idac.c` The Bemis–Murcko scaffold split is produced by `scripts/prepare_data.py`. The controlled comparison (Fig. 2, §6.8) is produced by `scripts/experiments/run_e5_sigma_grounding.sh` and aggregated by `scripts/analysis/run_e5_comparison.py` over `results/e5_sigma_grounding/seed_{42,43,44}/`;

the 3-seed aggregate is tabulated in `results/e5_sigma_grounding/THREE_SEED_SUMMARY.md`. The closure–insufficiency decomposition (Table 1, Figs 3 and 4) is regenerated deterministically by `scripts/analysis/run_b_insuff_decomposition.py` → `results/b_insuff/decomposition.json` from the committed matched-pair table `results/b_insuff/matched_pairs.csv` (the n=60 VT-2005-matchable pairs). The model checkpoints and the full per-arm prediction CSVs are large and are deposited at [Zenodo DOI] rather than in git; the named scripts regenerate them from the raw sources. The manuscript compiles with `make` in `paper/`.

Supplementary Information

A Supplementary methods

Full graph featurisation and feature list; encoder/interaction hyperparameters; the three closures' exact parametrisations and COSMO-SAC constants (a_{eff}, α', hydrogen-bond cutoff, r_0, q_0); SLE solver iteration counts and the convergence audit; the complete per-phase loss taxonomy and weights. *[To be populated from `src/tgnn_solv/{model,layers,solver,loss,config}.py` and the training configs.]*

B Proofs

Proof of Lemma 1 (non-identifiability). Write $u = 1/T$ and $h = \Delta H_{\text{fus}}/R$, $u_m = 1/T_m$. At infinite dilution and with $\Delta C_p = 0$ the observable is

$$\mu(u) = -\Phi(u) - \ln\gamma_2^\infty(u) = -h(u-u_m) - (c + d\,u) = -(h+d)\,u + (hu_m - c),$$

using $\ln\gamma_2^\infty(u) = c + d\,u$ (affine in u by hypothesis). Thus μ depends on the parameters $\theta = (\Delta H_{\text{fus}}, T_m, c, d)$ only through the pair $(S,K) = (-(h+d),\, hu_m - c) \in \mathbb{R}^2$. The activity block spans this plane, $\partial(S,K)/\partial c = (0,-1)$ and $\partial(S,K)/\partial d = (-1,0)$; hence for *any* crystal value (h, u_m) and any target (S,K) there is an activity pair $d = -S-h$, $c = hu_m - K$ reproducing it. The likelihood level set therefore contains the full two-dimensional family $\{(h,u_m)\}$, so $(\Delta H_{\text{fus}}, T_m)$ are unidentified. Equivalently, the crystal score $\partial(S,K)/\partial h = (-1, u_m)$ lies in the span of the activity columns, so profiling out (c,d) annihilates it and $\mathcal{I}_{\text{eff}}(\Delta H_{\text{fus}}) = 0$. At finite dilution $\ln\gamma_2(u) = \ln\gamma_2^\infty(u)\,(1-x_2)^2$ adds $\ln\gamma_2^\infty(u)\,(2x_2 - x_2^2) = O(x_2)$, which is not affine in u because $x_2 = x_2(u)$; the crystal score then has a component outside the activity span of order x_2, so $\mathcal{I}_{\text{eff}}(\Delta H_{\text{fus}}) = O(x_2^2) > 0$. □

Proof of Lemma 2 (class-dependent information). This is the semiparametric efficient-score construction [14] (here finite-dimensional and Gaussian, so equivalently the partitioned-information / Frisch–Waugh–Lovell identity). With Gaussian noise of variance σ^2, parameter-of-interest score $v = \partial\mu/\partial\Delta H_{\text{fus}}$, and nuisance tangent space $\mathcal{T}_{\mathcal{H}}$ (the L^2-closure of the span of nuisance scores), the efficient score is the residual $\tilde{v} = v - \Pi_{\mathcal{H}}v$ and the efficient information is $\mathcal{I}_{\text{eff}} = \sigma^{-2}\|v - \Pi_{\mathcal{H}}v\|^2$. Here $v = \partial\mu/\partial\Delta H_{\text{fus}} = -(u-u_m)/R \in \text{span}\{1,u\}$. If $\mathcal{T}_{\mathcal{H}} \supseteq \text{span}\{1,u\}$ then $\Pi_{\mathcal{H}}v = v$ and $\mathcal{I}_{\text{eff}} = 0$, recovering Lemma 1; if $\mathcal{T}_{\mathcal{H}}$ is a proper subspace not containing v (e.g. a fixed slope, $\mathcal{T}_{\mathcal{H}} = \text{span}\{1\}$) then $\Pi_{\mathcal{H}}v \neq v$ and $\mathcal{I}_{\text{eff}} > 0$, with a finite Cramér–Rao bound. □

Proof of Lemma 3 (decomposition and bounds). (a) Identity. Split the residual as $m - g(z^\star) = [m - \mathbb{E}(m \mid z^\star)] + [\mathbb{E}(m \mid z^\star) - g(z^\star)]$, square, and take expectations. The squared terms give $\mathbb{E}[\text{Var}(m \mid z^\star)] = B_{\text{insuff}}$ and $\mathbb{E}[(\mathbb{E}(m \mid z^\star) - g(z^\star))^2] = B_{\text{closure}}$. The cross term vanishes: since

$\mathbb{E}(m \mid z^\star) - g(z^\star)$ is $\sigma(z^\star)$-measurable, the tower rule gives $\mathbb{E}\big[(m - \mathbb{E}(m \mid z^\star))(\mathbb{E}(m \mid z^\star) - g(z^\star))\big] = \mathbb{E}\big[(\mathbb{E}(m \mid z^\star) - g(z^\star))\,\mathbb{E}(m - \mathbb{E}(m \mid z^\star) \mid z^\star)\big] = 0$. *(b) Jensen lower bound.* With $\Delta(z^\star) = \mathbb{E}(m \mid z^\star) - g(z^\star)$, $B_{\text{closure}} = \mathbb{E}[\Delta^2] \geq (\mathbb{E}[\Delta])^2 = (\mathbb{E}[m] - \mathbb{E}[g])^2$ by Jensen, using $\mathbb{E}[\mathbb{E}(m \mid z^\star)] = \mathbb{E}[m]$; it requires only that m and g share a reference state, so that $\mathbb{E}[m] - \mathbb{E}[g]$ is a genuine mean discrepancy. *(c) Law-of-total-variance upper bound.* For any binning $B = \text{bin}(g(z^\star))$, $\sigma(B) \subseteq \sigma(z^\star)$, so conditioning on the coarser σ-algebra cannot reduce expected conditional variance: $\mathbb{E}[\text{Var}(m \mid B)] \geq \mathbb{E}[\text{Var}(m \mid z^\star)] = B_{\text{insuff}}$. *(d) Convention-independence.* $B_{\text{insuff}} = \mathbb{E}[\text{Var}(m \mid z^\star)]$ is a functional of the law of $(m, z^\star)$ alone and does not involve g; hence it is identical across closure conventions, and any $\text{MSE}(g_1) - \text{MSE}(g_2)$ is entirely $B_{\text{closure}}(g_1) - B_{\text{closure}}(g_2)$. $\square$

C Per-arm tables and decomposition robustness

Table 5 gives the exact per-arm metrics behind Fig. 2: three seeds (mean±sd) on the cross-arm intersection lock (n=5608, rows supervised and finite in every arm). Both closure families—NRTL and COSMO-SAC over a learned σ-profile—trail the DirectGNN control (the physics tax); the σ-oracle (true VT-2005 profile) is the worst arm, and restoring the Staverman–Guggenheim combinatorial term makes it slightly worse still—the signature of a misspecified map (§6.2).

Table 5: Per-arm scaffold-split metrics (3 seeds, n=5608 cross-arm lock). Ordered by MAE; the physics closures trail DirectGNN and the true-σ oracle is worst. Regenerated by `run_e5_comparison.py` over `results/e5_sigma_grounding/seed_{42,43,44}/`.

Arm	MAE	R^2
DirectGNN (no closure)	1.70 ± 0.03	$+0.42 \pm 0.03$
NRTL closure	1.80 ± 0.07	$+0.34 \pm 0.03$
COSMO-SAC, learned σ (grounded)	1.85 ± 0.05	$+0.33 \pm 0.05$
+ Staverman–Guggenheim combinatorial	1.88 ± 0.09	$+0.32 \pm 0.05$
COSMO-SAC, ungrounded σ	2.04 ± 0.04	$+0.19 \pm 0.01$
COSMO-SAC, true σ (σ-oracle)	2.25 ± 0.02	-0.03 ± 0.04

The closure/insufficiency decomposition (full/res conventions; LOTV, RF, ridge, and kNN estimators; solvent-clustered bootstrap; the pyridine stratum; the four hedges) is tabulated in §6.2 and regenerated by `run_b_insuff_decomposition.py`.

D External baselines and ablations

We retrained the direct SOTA (FastSolv [1]) and the physics-informed SOTA (SolProp [15]) on our corpus and evaluated them on a by-solute test split (`test_solute.csv`, $n \approx$ 10–12k). In $\ln x_2$ they reach MAE 1.94 (FastSolv, R^2 0.23) and 2.34 (SolProp, calibrated, R^2 −0.18); in $\log_{10} S$, 0.87 and 1.04. Neither beats our matched arms, and the physics-informed SolProp trails the direct FastSolv—the same ordering we find internally between the physics closures and DirectGNN. These runs use a different (by-solute) test file than the scaffold split of the headline comparison, so they bound the task's difficulty rather than rank directly against the scaffold numbers of Table 5; a like-for-like scaffold evaluation of the external models is left to the deposited artifacts. The regime-level reading is that on this corpus no black box, ours or external, escapes the accuracy ceiling—consistent with the structural account of §7. Per-arm ablations (NRTL vs COSMO-SAC, combinatorial on/off) are in Table 5.

E Supplementary mechanism results

The corrupted-twin non-identifiability demonstration; crystal grounding on physical labels (T_m MAE 45 K); the omitted-heat-capacity (ΔC_p) audit; fusion-supervision scarcity; the NRTL-branch internal-collapse forensics; the UMAP ΔMAE cluster map.

F Data provenance and reproducibility

BigSolDB / VT-2005 / ThermoML provenance; split construction and the °C/K trap; seeds; the proxy-versus-GPU compute ledger for every reported number.

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